

# DEVANSH SHUKLA

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## SUMMARY

Pre-final year Computer Science undergraduate specializing in **LLM-based systems** and **multi-agent decision architectures**, with experience building **scalable backend systems** using **FastAPI** and **MongoDB**. Strong algorithmic foundation through competitive programming, with focus on designing **explainable, risk-aware AI systems**.

## EDUCATION

**Indian Institute Of Information Technology, Bhopal**

Sept 2023 – Jun 2027

B.Tech. in Computer Science & Engineering

**CGPA: 8.37**

Relevant Coursework: **Data Structures & Algorithms, Object-Oriented Programming**

## EXPERIENCE

**Research Project | Hardware Optimization for DSP & Image Processing Systems**

Jan 2026 – Present

- Optimized fault-tolerant hardware scheduling using Particle Swarm Optimization (PSO) and **DMR/TMR** techniques, reducing execution latency by **25%** and improving area efficiency by **18%** across DSP benchmarks.

**Team Leader | Agentic AI Project, EY Techathon (Round 2 Qualifier)**

Dec 2025 – Jan 2026

- Placed among the **top 4,000 teams** out of approximately **2 lakh registered participants**, leading a **5-member team** to build **Stratabid™**, an **agentic RFP orchestration system** with **4 parallel agent workflows**.
- Implemented a **hybrid matching engine** using **FAISS vector search** and **weighted rule-based scoring** (Voltage **35%**, Material **30%**), applied across **100+ specification parameters**, targeting up to **80% reduction in bid turnaround time (TAT)**.

**Executive | CODAME, IIIT Bhopal**

Nov 2024 – Dec 2025

- Spearheaded planning and execution of coding contests and workshops, engaging over 550 students and driving a 50% increase in participation through strategic outreach to 10+ institutes.
- Executed flagship events including FLOC and Reverse Coding Contest, optimizing logistics, volunteer coordination, and resource allocation; streamlined operations reduced setup time by 30%, enhancing participant experience.

## PROJECTS

**Courtroom Trading System** | Python, FastAPI, MongoDB

[GitHub](#) [Live Site](#)

- Architected a **multi-agent LLM-based decision system** (Bull–Bear–Judge), enabling **100% explainable outputs** with structured reasoning across agents.
- Designed a **hybrid reasoning pipeline** combining LLM outputs with deterministic validation using **four key signals**: momentum, trend alignment, ATR-based volatility, and risk-reward constraints.
- Developed a **persistent reasoning layer** in MongoDB storing **4+ fields per trade** (bullish/bearish factors, reasoning, setup), reducing repeat inference to **0 additional LLM calls** and enabling **O(1)** retrieval for history replay.

**GlidePAY** | Flask, MongoDB, React.js, REST APIs

[GitHub](#) [Live Site](#)

- Built a custom **UPI-style** payment app with **Flask** and **MongoDB**, enabling QR-based **peer-to-peer** transfers and multi-mode payments for **20+ active users**.
- Ensured **100% ACID compliance** using MongoDB **transactions** and implemented **session-based authentication** for secure user interactions.
- Deployed the front-end (**React.js**) and back-end (**Flask**) via Render with seamless **REST API integration**; reduced API latency by **35%** through efficient routing improvements.

## TECHNICAL SKILLS

- Languages:** Python, C++, Java, JavaScript, SQL
- Frameworks:** FastAPI, Flask, React.js, Jinja2, Bootstrap
- Databases:** PostgreSQL, MongoDB, SQLite
- Developer Tools:** Git, GitHub, Linux, Postman
- Core Areas:** LLM-based Systems, Multi-Agent Systems, Applied AI Decision Systems

## ACHIEVEMENTS

- Excelled in competitive programming: secured CodeChef **4-star (1824)**, Codeforces **Specialist (1506)**, and LeetCode **Knight (1902)** while conquering **300+** algorithmic challenges spanning data structures, algorithms, and algorithmic optimization.
- Dominated global competitive programming contests: secured top ranks on CodeChef (**180th, 336th, 406th of 28,000+**), Codeforces (**608th of 40,000+**), and LeetCode (**321st of 30,000+**), showcasing exceptional algorithmic skill under high-pressure scenarios.
- Semi-Finalist, Tata Imagination Challenge:** Ranked among the **Top 5,000** participants out of **10 lakh+** nationwide applicants in a national-level strategy and innovation competition.